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T12 – AI&DS

SUB : ASD&D Lab

EXPERIMENT – 1

AIM : To perform version control operations using Git.

LO MAPPED : LO2.

LO2 : Use git effectively for version control, source code management and collaborative team development.

THEORY :

1. Starting a Repository

- git init — Initializes a brand new, local Git repository in your current directory.
- git clone <url> — Copies an existing remote repository (from GitHub, GitLab, etc.) onto your local machine.

2. Configuring Git

- git config --global user.name "Your Name" — Sets your username globally. This name appears as the author of every commit you make.
- git config --global user.email "your@email.com" — Sets your email address globally. This email is associated with your commits and helps identify you in the repository history.

3. Making and Tracking Changes

- git status — Shows the current state of your working directory (which files are modified, staged, or untracked).
- git add <file> — Stages a specific file, preparing it to be included in your next commit.
- git add . — Stages all modified and untracked files in the current directory at once.
- git commit -m "message" — Saves your staged snapshot to the project history with a descriptive message.
- git commit --amend -m "new message" — Modifies your very last commit (great for fixing a typo in your last commit message).

4. Viewing History & Differences

- git log — Displays the commit history for the current branch, showing commit IDs, authors, dates, and messages.
- git log --oneline — Shows a highly condensed, one-line version of your commit history for quick scanning.
- git diff — Shows the exact line-by-line differences between your current working directory and your last save.
- git diff --staged — Shows the differences between your staged files and your last commit.
- git blame <file> — Displays file line-by-line, showing exactly who made the last change to each line and in which commit.

5. Branching & Merging

- git branch — Lists all local branches in your repository. The current active branch will have an asterisk (*).
- git branch <branch-name> — Creates a new branch at your current commit but doesn't switch to it.

- git checkout <branch-name> — Switches your working directory to the specified branch.
- git switch -c <branch-name> — A modern, dedicated command that creates a new branch and instantly switches to it.
- git merge <branch-name> — Integrates the history from the specified branch into your *current* active branch.
- git branch -d <branch-name> — Deletes a local branch, safely preventing you from deleting it if it has unmerged changes.

6. Working with Remotes

- git remote add origin <url> — Links your local repository to a remote server (naming it "origin").
- git push -u origin <branch-name> — Uploads your local branch commits to the remote repository and sets it as the default tracking branch.
- git push — Uploads your local commits to the remote repository (once the tracking branch is set).
- git pull — Fetches updates from the remote repository and instantly merges them into your current local branch.
- git fetch — Downloads history and changes from the remote repository *without* modifying your local working files (safer than a pull if you want to inspect first).

git stash — Temporarily saves all your uncommitted changes (both staged and unstaged) and restores your working directory to the last committed state.

git rm <file> — Removes a tracked file from both the working directory and the Git repository. The deletion is staged automatically.

OUTPUT :

```
C:\Users\203\Desktop\GIT_DEMO>git --version
git version 2.43.0.windows.1

C:\Users\203\Desktop\GIT_DEMO>git clone https://github.com/Aryan-Kashid/AgileSoftware.git
Cloning into 'AgileSoftware'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.

C:\Users\203\Desktop\GIT_DEMO>git status
On branch main
Changes not staged for commit:
  (use "git add/rm <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        deleted:    ../README.md
```

```
C:\Users\203\Desktop\GIT_DEMO>cd AgileSoftware

C:\Users\203\Desktop\GIT_DEMO\AgileSoftware>git add index.html
```

```
C:\Users\203\Desktop\GIT_DEMO\AgileSoftware>git config --global user.name "Aryan Kashid"

C:\Users\203\Desktop\GIT_DEMO\AgileSoftware>git config --global user.email "aryankashid985@gmail.com

C:\Users\203\Desktop\GIT_DEMO\AgileSoftware>git commit -m "ADDED NEW FILE"
[main 616f755] ADDED NEW FILE
 1 file changed, 12 insertions(+)
 create mode 100644 index.html

C:\Users\203\Desktop\GIT_DEMO\AgileSoftware>git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean

C:\Users\203\Desktop\GIT_DEMO\AgileSoftware>git push origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 20 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 461 bytes | 461.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/Aryan-Kashid/AgileSoftware.git
   e952c56..616f755  main -> main
```

```
C:\Users\203\Desktop\GIT_DEMO\AgileSoftware>cd ..

C:\Users\203\Desktop\GIT_DEMO>mkdir DEMO_2

C:\Users\203\Desktop\GIT_DEMO>cd DEMO_2

C:\Users\203\Desktop\GIT_DEMO\DEMO_2>git init
Initialized empty Git repository in C:/Users/203/Desktop/GIT_DEMO/DEMO_2/.git/

C:\Users\203\Desktop\GIT_DEMO\DEMO_2>cd ..

C:\Users\203\Desktop\GIT_DEMO>cd AgileSoftware

C:\Users\203\Desktop\GIT_DEMO\AgileSoftware>git commit --amend -m "NEW CHANGES"
[main 899d4fa] NEW CHANGES
 Date: Tue Jul 14 10:25:59 2026 +0530
 1 file changed, 12 insertions(+)
 create mode 100644 index.html

C:\Users\203\Desktop\GIT_DEMO\AgileSoftware>git log
commit 899d4fa0bcd74cb3c426195b658c82754272d1c7 (HEAD -> main)
Author: Aryan Kashid <aryankashid985@gmail.com>
Date:   Tue Jul 14 10:25:59 2026 +0530

    NEW CHANGES

commit e952c561f89350c87045625c601cca9730482eb2
Author: Aryan-Kashid <aryankashid985@gmail.com>
Date:   Tue Jul 14 10:12:19 2026 +0530

    Initial commit
```

```
C:\Users\203\Desktop\GIT_DEMO\AgileSoftware>git log --oneline
899d4fa (HEAD -> main) NEW CHANGES
e952c56 Initial commit

C:\Users\203\Desktop\GIT_DEMO\AgileSoftware>git branch
* main

C:\Users\203\Desktop\GIT_DEMO\AgileSoftware>git branch -M main

C:\Users\203\Desktop\GIT_DEMO\AgileSoftware>git checkout -b features
Switched to a new branch 'features'

C:\Users\203\Desktop\GIT_DEMO\AgileSoftware>git checkout -b demo
Switched to a new branch 'demo'

C:\Users\203\Desktop\GIT_DEMO\AgileSoftware>git branch
* demo
  features
  main

C:\Users\203\Desktop\GIT_DEMO\AgileSoftware>git checkout demo
Already on 'demo'
```

```
C:\Users\203\Desktop\GIT_DEMO\AgileSoftware>git checkout main
Switched to branch 'main'
Your branch and 'origin/main' have diverged,
and have 1 and 1 different commits each, respectively.
  (use "git pull" if you want to integrate the remote branch with yours)

C:\Users\203\Desktop\GIT_DEMO\AgileSoftware>git branch -d demo
Deleted branch demo (was 899d4fa).

C:\Users\203\Desktop\GIT_DEMO\AgileSoftware>git branch
  features
* main
```

```
C:\Users\203\Desktop\GIT_DEMO\AgileSoftware>git diff main
diff --git a/app.js b/app.js
new file mode 100644
index 0000000..e69de29

C:\Users\203\Desktop\GIT_DEMO\AgileSoftware>git merge main
Already up to date.
```

```
C:\Users\203\Desktop\GIT_DEMO>cd AgileSoftware

C:\Users\203\Desktop\GIT_DEMO\AgileSoftware>git checkout main
Already on 'main'
Your branch and 'origin/main' have diverged,
and have 2 and 1 different commits each, respectively.
  (use "git pull" if you want to integrate the remote branch with yours)

C:\Users\203\Desktop\GIT_DEMO\AgileSoftware>git pull origin main
From https://github.com/Aryan-Kashid/AgileSoftware
 * branch            main       -> FETCH_HEAD
Merge made by the 'ort' strategy.

C:\Users\203\Desktop\GIT_DEMO\AgileSoftware>git merge features
Already up to date.

C:\Users\203\Desktop\GIT_DEMO\AgileSoftware>git push origin main
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 262 bytes | 262.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/Aryan-Kashid/AgileSoftware.git
   616f755..9f48aa8  main -> main

C:\Users\203\Desktop\GIT_DEMO\AgileSoftware>git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean

C:\Users\203\Desktop\GIT_DEMO\AgileSoftware>
```

```
C:\Users\203\Desktop\GIT_DEMO\AgileSoftware>git blame index.html
899d4fa0 (Aryan Kashid 2026-07-14 10:25:59 +0530  1) <!DOCTYPE html>
899d4fa0 (Aryan Kashid 2026-07-14 10:25:59 +0530  2) <html lang="en">
899d4fa0 (Aryan Kashid 2026-07-14 10:25:59 +0530  3) <head>
899d4fa0 (Aryan Kashid 2026-07-14 10:25:59 +0530  4)     <meta charset="UTF-8">
899d4fa0 (Aryan Kashid 2026-07-14 10:25:59 +0530  5)     <meta name="viewport" content="width=device-width, initial-scale=1.0">
899d4fa0 (Aryan Kashid 2026-07-14 10:25:59 +0530  6)     <title>git Demo</title>
899d4fa0 (Aryan Kashid 2026-07-14 10:25:59 +0530  7) </head>
899d4fa0 (Aryan Kashid 2026-07-14 10:25:59 +0530  8) <body>
899d4fa0 (Aryan Kashid 2026-07-14 10:25:59 +0530  9)     <h1>Version Control Operations</h1>
899d4fa0 (Aryan Kashid 2026-07-14 10:25:59 +0530 10)
899d4fa0 (Aryan Kashid 2026-07-14 10:25:59 +0530 11) </body>
899d4fa0 (Aryan Kashid 2026-07-14 10:25:59 +0530 12) </html>

C:\Users\203\Desktop\GIT_DEMO\AgileSoftware>
```

```
● aryankashid@ARYANs-MacBook-Air GIT_DEMO % cd AgileSoftware
● aryankashid@ARYANs-MacBook-Air AgileSoftware % git diff HEAD
● aryankashid@ARYANs-MacBook-Air AgileSoftware % git diff --staged
```

```
● aryankashid@ARYANs-MacBook-Air GIT_DEMO % mkdir demo_project
● aryankashid@ARYANs-MacBook-Air GIT_DEMO % cd demo_project
● aryankashid@ARYANs-MacBook-Air demo_project % git init
  Initialized empty Git repository in /Users/aryankashid/Desktop/GIT_DEMO/demo_project/.git/
● aryankashid@ARYANs-MacBook-Air demo_project % git status
  On branch main

  No commits yet

  nothing to commit (create/copy files and use "git add" to track)
● aryankashid@ARYANs-MacBook-Air demo_project % echo "# Demo Project" > README.md
● aryankashid@ARYANs-MacBook-Air demo_project % git add README.md
● aryankashid@ARYANs-MacBook-Air demo_project % git commit -m "Initial commit"
  [main (root-commit) 9ed3927] Initial commit
   1 file changed, 1 insertion(+)
   create mode 100644 README.md
● aryankashid@ARYANs-MacBook-Air demo_project % git remote -v
● aryankashid@ARYANs-MacBook-Air demo_project % git remote add origin https://github.com/Aryan-Kashid/demo_project.git
● aryankashid@ARYANs-MacBook-Air demo_project % git remote -v
  origin  https://github.com/Aryan-Kashid/demo_project.git (fetch)
  origin  https://github.com/Aryan-Kashid/demo_project.git (push)
● aryankashid@ARYANs-MacBook-Air demo_project % git push -u origin main
  Enumerating objects: 3, done.
  Counting objects: 100% (3/3), done.
  Writing objects: 100% (3/3), 232 bytes | 232.00 KiB/s, done.
  Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
  To https://github.com/Aryan-Kashid/demo_project.git
   * [new branch]      main -> main
  branch 'main' set up to track 'origin/main'.
```

```
● aryankashid@ARYANs-MacBook-Air demo_project % git remote
  origin
● aryankashid@ARYANs-MacBook-Air demo_project % git remote show origin
  * remote origin
    Fetch URL: https://github.com/Aryan-Kashid/demo_project.git
    Push  URL: https://github.com/Aryan-Kashid/demo_project.git
    HEAD branch: main
    Remote branch:
      main tracked
    Local branch configured for 'git pull':
      main merges with remote main
    Local ref configured for 'git push':
      main pushes to main (up to date)
● aryankashid@ARYANs-MacBook-Air demo_project % git fetch origin
● aryankashid@ARYANs-MacBook-Air demo_project % git remote rename origin github
  Renaming remote references: 100% (3/3), done.
● aryankashid@ARYANs-MacBook-Air demo_project % git remote remove github
```

```
● aryankashid@ARYANs-MacBook-Air AgileSoftware % git stash
  Saved working directory and index state WIP on main: 9f48aa8 Merge branch 'main' of https://github.com/Aryan-Kashid/AgileSoftwa
● aryankashid@ARYANs-MacBook-Air AgileSoftware % git reset --hard
  HEAD is now at 9f48aa8 Merge branch 'main' of https://github.com/Aryan-Kashid/AgileSoftware
⊗ aryankashid@ARYANs-MacBook-Air AgileSoftware % git rm
  fatal: No pathspec was given. Which files should I remove?
● aryankashid@ARYANs-MacBook-Air AgileSoftware % git rm app.js
  rm 'app.js'
```

CONCLUSION :

This experiment provided valuable hands-on experience with core Git commands, effectively bridging the gap between local development and remote version control on GitHub. By exploring operations like branching, merging, and remote repository management, I have gained a practical understanding of how to maintain code history and collaborate efficiently in a professional development environment.